

B12 deficiency in adults

1: Presentation and investigation

NICE NG239, 2024, BMJ 2023;383:e071725, Mayo Clin Proc Innov Qual Outcomes 2019;3:200



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When to test?

1 or more common symptoms/signs of B12 deficiency



1 or more common risk factors for B12 deficiency

Test for B12 deficiency

1 or more common symptoms/signs of B12 deficiency

BUT no common risk factors

Use clinical judgement to decide whether to test for B12 deficiency

The judgement is around: likelihood of an unrecognised/undiagnosed risk factor (e.g. undiagnosed atrophic gastritis) OR are the clinical features such that missing B12 deficiency as a cause could be more devastating, e.g. ataxia.

Common symptoms and signs

- **Blood:** anaemia or macrocytosis.
N.B. do not rule out B12 deficiency if Hb +/- MCV are normal (macrocytic anaemia occurs in <20%).
- **Balance** or neurological problems:
 - Balance problems, impaired gait or falls.
 - Sensory ataxia or paraesthesia.
- **Brain:** cognitive problems:
 - Poor concentration.
 - Short-term memory loss/brain fog.
- **Tiredness:** unexplained fatigue.
- **Tongue:** glossitis.
- **Optic nerve dysfunction:**
 - Blurred vision or optic atrophy.
 - Visual field loss (scotoma).
- **Iron:** poor response to iron replacement in pregnancy/breastfeeding.

NICE identified an association between mental health issues and B12 deficiency, BUT there was a lack of evidence to recommend testing.

Common risk factors

- **Diet** low in B12 (and not taking OTC B12):
 - No/infrequent animal-source foods.
 - No food/drinks fortified with B12.
 - Allergy to eggs, milk or fish.
 - Poor access to/inability to prepare foods containing B12, e.g. dementia, mental health, low income.
 - Restricted diet, e.g. eating disorder.
- **Disease/intervention:**
 - Any autoimmune condition, e.g. coeliac, thyroid disease, type 1 diabetes.
 - Atrophic gastritis of the gastric body.
 - Gastrointestinal surgery, e.g. gastrectomy, terminal ileum resection, bariatric surgery (Roux-en-Y bypass or sleeve gastrectomy).
 - Abdominal or pelvic radiotherapy.
- **Drugs:** e.g. metformin, colchicine, PPIs, H2-receptor antagonists, phenobarbital, pregabalin, primidone, topiramate, recreational nitrous oxide use.
- **Family history:**
 - Of B12 deficiency.
 - Of any autoimmune condition.

Which test?

In certain situations, do not wait on blood results. Immediate replacement is required. See p2 for details.

There are 4 tests:

- Total B12 (serum cobalamin).

- Active B12 (serum holotranscobalamin).

- Plasma homocysteine (needs to be taken in secondary care; sample directly to lab on ice).
- Serum methylmalonic acid (MMA).

Most labs report total B12. NICE recommends:

Measure total B12 (serum cobalamin) or active B12 (serum holotranscobalamin).

Exceptions:

- If pregnant: always measure serum holotranscobalamin.

- If nitrous oxide use: **measure plasma homocysteine or serum methylmalonic acid (MMA).**

When measuring total B12 or active B12, NICE highlights 2 issues:

- Taking OTC B12 can cause rises in total/active B12 but mask a deficiency that has not been properly addressed. *We think this means the following: for example, a patient develops glossitis, reads that B12 deficiency is a cause and starts OTC oral B12. Bloods then show a normal B12 level, but the cause of the glossitis IS a (partially treated) B12 deficiency (risking the clinician being falsely reassured by the normal B12 result).*
- COCP use: COCP can lower total B12 levels without causing a deficiency. In practice, it is then difficult to know whether this is a true B12 deficiency or a false positive.



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2: Test results and management

NICE NG239, 2024, British Journal of Haematology 2014;166:496



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In certain situations, treatment should be commenced/considered immediately, without waiting for test results:

Suspected megaloblastic anaemia AND neurological symptoms (especially any symptoms of subacute combined degeneration of the spinal cord)

What is a 'suspected megaloblastic anaemia'? Symptoms/signs of B12 deficiency with or without an abnormal FBC; see article for full details (Red Whale correspondence with NICE, December 2024)

Start B12 replacement immediately (see p3) of GEMS

Test B12, DO NOT WAIT FOR TEST RESULTS before starting treatment

Suspected B12 deficiency and cause is likely to be due to a drug

CONSIDER starting B12 replacement while awaiting test results

For those unaffected by the above conditions:

Interpreting test results:

- **Total B12/holotranscobalamin:** use reference ranges below, UNLESS local lab reports substantially different ranges, when local lab thresholds should be used.
- **Homocysteine:** use lab reference range (rises in B12 deficiency; also raised if low folate).
- **MMA:** use lab reference range (rises in B12 deficiency).

Most labs report total or active B12....use the right one!

	Total B12 (commonest test)	Holotranscobalamin (active B12)
B12 deficiency	<180ng/L or <133pmol/L	<25pmol/L
Indeterminate level: possible B12 deficiency	180–350ng/L or 133–258pmol/L	25–70pmol/L
Normal	>350ng/L or >258pmol/L	>70pmol/L

Confirmed B12 deficiency

Indeterminate level (possible B12 deficiency)

Normal result (B12 deficiency unlikely)

People of black ethnicity may have a higher reference range for B12. Treatment may be needed even if blood results are above cut-offs; consider symptoms, signs and risk factors.

B12 REPLACEMENT NEEDED (see p3).

START B12 treatment AND look for cause: is test for anti-intrinsic factor antibody indicated?

(labs call this intrinsic factor antibody)

Pregnant/breastfeeding

Test for anti-intrinsic factor antibody

Previous positive anti-intrinsic factor antibody OR previous operation which could cause B12 malabsorption, e.g. total gastrectomy/terminal ileum resection

Testing for anti-intrinsic factor antibody won't change management so is not needed

Everyone else AND if cause of B12 deficiency unknown

CONSIDER testing for anti-intrinsic factor antibody and offer coeliac serology

May need: B12 replacement OR better test OR no action

(We have put what we think is NICE's rationale in italics)
If decide to treat, manage as for confirmed deficiency (p3)

Pregnant/breastfeeding

Underlying neurological/haematological condition that could deteriorate rapidly
e.g. anaemia or ataxia

Underlying irreversible cause
e.g. autoimmune gastritis, gastrectomy, terminal ileum resection, certain bariatric surgery (e.g. Roux-en-Y bypass or sleeve gastrectomy)

Symptoms/signs of B12 deficiency BUT none of the above conditions

NO symptoms/signs of B12 deficiency (and none of the above)

Look for alternative cause. If ongoing symptoms after 3–6m, consider repeating B12

Consider treating (because of potential risks of not treating?)
MMA may be helpful to confirm, but treat while awaiting result

Consider treating because at high risk of developing B12 deficiency (all should be on B12 prophylaxis so MMA unlikely to add value)

Consider doing a better test (MMA) to get more accurate idea of B12 status

Watch and wait: review if symptoms develop

Anti-intrinsic factor antibody results: if positive, autoimmune gastritis is likely. A negative result doesn't rule out autoimmune gastritis (sensitivity 40–60%). If anti-IF negative and autoimmune gastritis still suspected, consider one of: anti-gastric parietal cell antibody, gastrin level, cobaSorb test or OGD with biopsy.

B12 deficiency in adults

3: Management

NICE NG239, 2024, BMJ 2023;383:e071725, NEJM 2013;368:149, BNF, SPC hydroxocobalamin, Derbyshire shared care pathology guidelines vitamin B12 and folate, all accessed May 2024, BJH 2014;166:496, British Obesity and Metabolic Surgery Society (2020)



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- Response to treatment varies depending on the underlying cause.
- Symptoms may initially worsen. Symptoms should start to improve within 2w, but may take up to 3m. Complete resolution can be much longer (potentially months to years for neurological symptoms).
- Anaemia, if present, usually resolves within 6–8w of starting treatment.
- Review if symptoms have not improved, worsen, return or new symptoms appear after starting treatment.

Management depends on the underlying cause...

IM B12 must be used if:

- Autoimmune gastritis.
- Total gastrectomy or complete terminal ileum resection.
- Pregnant/breastfeeding and autoimmune gastritis suspected.

In other situations, oral or IM can be used.

What if neurological symptoms at presentation? The BNF outlines an increased frequency of IM replacement for loading AND maintenance if neurological symptoms (*we would suggest the safest option would be IM replacement*).

Cause	Route	Duration and notes
Malabsorption from autoimmune gastritis, total gastrectomy or complete terminal ileum resection	IM	Duration: lifelong. Intramuscular B12 must be used.
Malabsorption: other causes , e.g. coeliac disease, partial gastrectomy or other bariatric surgery	Consider IM	Duration: depends on whether cause treatable or not (see p4 for details). Why IM? More cost-effective if used ≥6m. Oral absorption difficult to judge. If using oral, suggested B12 dose ≥1mg/d cyanocobalamin.
Medication-induced (note MHRA guidance on metformin in main article)	Oral or IM	Duration: while using causative drug and until all symptoms resolved. No evidence oral/IM superior to the other. Review the causative medication – can this be stopped/changed? If causative drug stopped and no ongoing symptoms: review need for B12.
Recreational nitrous oxide	Oral or IM	Duration: while using causative drug and until all symptoms resolved. Advise stopping use of nitrous oxide. If stops, review need for B12.
Diet-related (Lots of oral B12 supplements are available OTC: NICE recommends using one of: cyanocobalamin, methylcobalamin or adenosylcobalamin)	Consider oral. OTC for most	Duration: as long as diet remains unchanged and until symptoms resolved. Consider alternative causes (may be multifactorial); consider further tests. Ask about diet/drinks and OTC B12 (see p1 for rationale). Consider IM B12 if: • Another condition that may deteriorate rapidly/have major effect on life, e.g. ataxia/anaemia. • Concerns about adherence to oral treatment, e.g. multimorbidity, frailty, cognitive impairment or homeless.
Unknown cause (provided not likely to be malabsorption)	Consider oral	Duration: likely will need to continue. Cause unknown so cannot be addressed.

What dose? (based on BNF)

Current UK practice is to use IM hydroxocobalamin. For IM cyanocobalamin doses, see BNF.
Where the BNF says 'pernicious anaemia' (a term NICE avoids), we use the phrase 'B12 deficiency'.

Intramuscular B12 (hydroxocobalamin)

B12 deficiency with neurological involvement	• Initially: 1mg hydroxocobalamin IM on alternate days until symptoms stop improving, then swap to maintenance. • Maintenance: 1mg every 2 months.
B12 deficiency without neurological involvement	• Initially: 1mg hydroxocobalamin IM 3 times a week, for 2 weeks. • Maintenance: 1mg IM every 2–3 months.
Prophylaxis of B12 deficiency, e.g. in those who have had gastrectomy, ileal resection (BNF)	• 1mg hydroxocobalamin IM every 2–3 months (BNF). • Obesity and Metabolic Surgery Society: for sleeve gastrectomy and Roux-en-Y, give 1mg hydroxocobalamin IM every 3m.

ORAL B12 (cyanocobalamin): BUT see top of page for certain times when oral replacement should **NOT** be used

Due to malabsorption (when IM replacement isn't required)	NICE recommends ≥1mg/day (off-licence).
Dietary deficiency	In pregnancy/breastfeeding: 1mg/d. Everyone else: 50–150mcg/d.
Other causes (when IM replacement isn't required)	NICE/BNF do not comment on dosing. British Journal of Haematology suggests cyanocobalamin 1–2mg/d (off-licence).

NB: if deficient in both folate and B12, start B12 replacement at least 24h before starting folate ('B before F')

B12 deficiency in adults

4: Ongoing management and troubleshooting

NICE NG239, 2024, BNF, accessed May 2024, BMJ 2023;383:e071725, SPC hydroxocobalamin
https://www.medicines.org.uk/emc/product/4662/smpc#ref, accessed May 2024, Gut 2019;68:1545



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Follow-up for those on IM or oral B12

If pregnancy/breastfeeding: offer initial follow-up at **1m**.

Everyone else: offer initial follow-up 3m after starting treatment, or earlier depending on symptom severity.

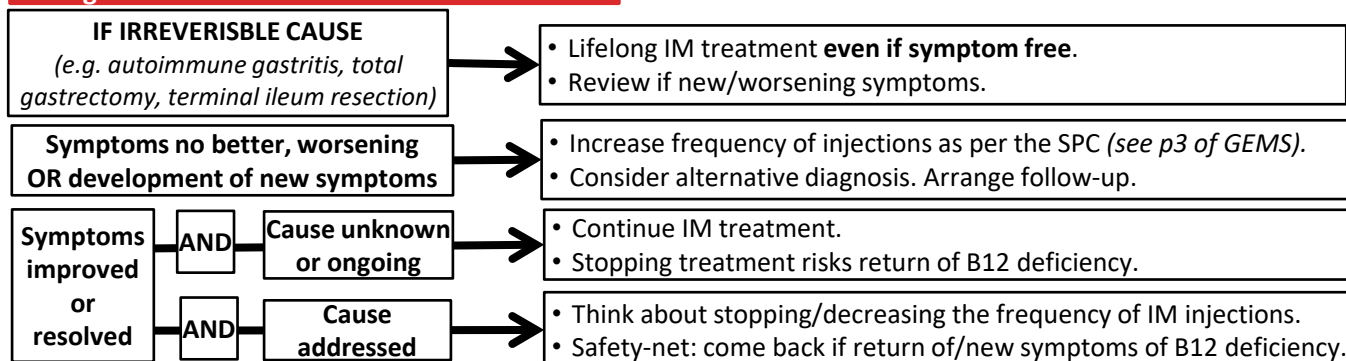
When to stop treatment?

For many, treatment is lifelong because the underlying cause is irreversible.

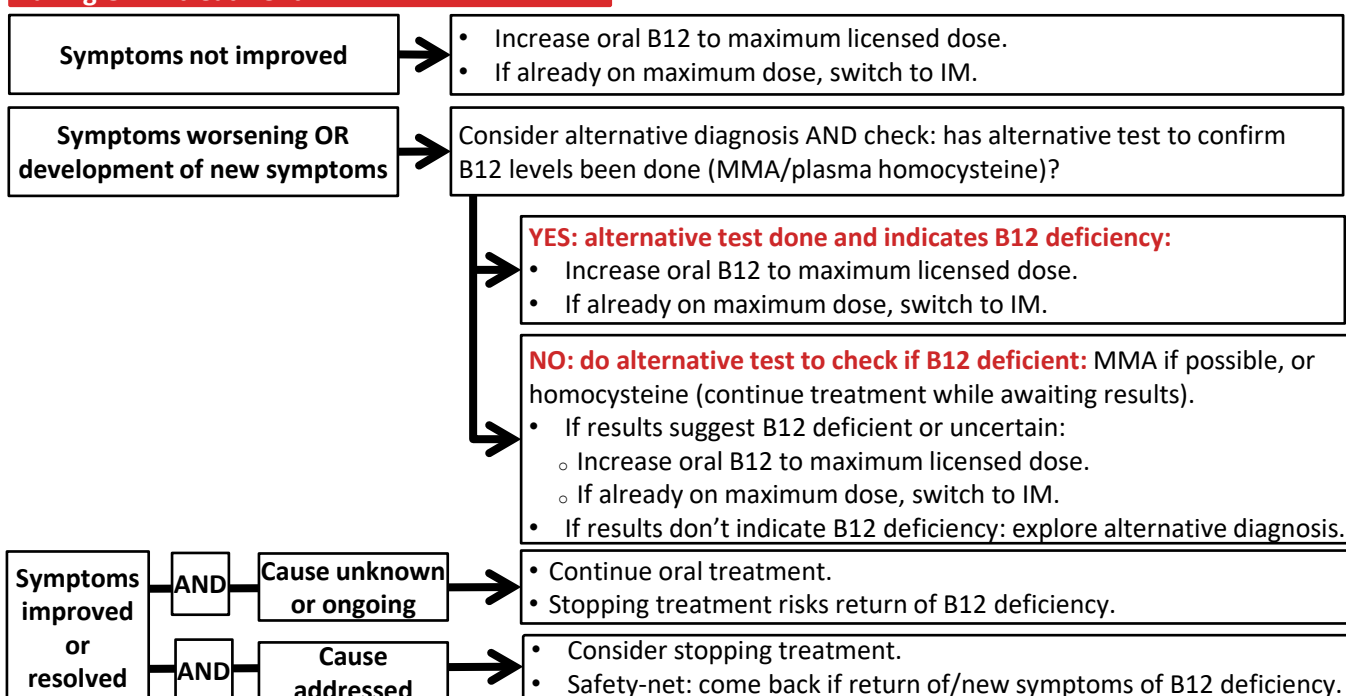
Treatment can be stopped if all symptoms resolve AND underlying cause is identified AND resolved, e.g. stopped being vegetarian; coeliac disease was the cause and stable on strict gluten-free diet.

If appropriate to stop but pregnant/breastfeeding, continue until delivery/stops breastfeeding (benefits of treating outweigh risk of stopping).

Taking INTRAMUSCULAR treatment



Taking ORAL treatment



Should B12 levels be rechecked when on IM B12 treatment?

No, not for those on IM therapy. Treatment will impact the test results so this is not useful in practice; this is the case for both B12 levels and MMA results. Blood test results should not be used to monitor clinical improvement or decide on the frequency of intramuscular B12. Instead, be guided by the presence, or resolution, of symptoms.

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